

**Automatic speech recognition for Moroccan
Darija and Tarifit:
Optimizing OpenAI Whisper for a resource-
constrained dialect**



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1. Abstract

This study investigates the adaptation of OpenAI's Whisper models for Automatic Speech Recognition (ASR) in two under-resourced Moroccan languages: Moroccan Darija and Tarifit. For Darija, fine-tuning the Whisper Small model on approximately 4,500 speech samples reduced the Word Error Rate (WER) from 133.4% to 45.8%, corresponding to a relative improvement of 65.7%. For Tarifit, the lack of publicly available speech resources necessitated the creation of a dedicated corpus consisting of 120 manually transcribed audio samples. Initial evaluations revealed very high transcription error rates across all Whisper variants, with Whisper Large-v3 achieving the strongest baseline performance. However, Whisper Small was selected for adaptation due to its favorable balance between accuracy and computational requirements. After fine-tuning, the WER decreased from 114.58% to 88.75%, representing a relative improvement of 22.55%. The experiments, conducted using the Hugging Face Transformers framework on NVIDIA T4 GPUs, demonstrate that transfer learning and task-specific fine-tuning can substantially improve ASR performance for low-resource Moroccan languages. Furthermore, the study highlights the critical role of speech corpus development in advancing speech recognition research for dialectal Arabic and Amazigh language varieties.

2. Keywords:

Automatic Speech Recognition; Moroccan Darija; Whisper; Fine-tuning; Word Error Rate; Low-resource languages; Arabic dialects; Transfer learning; Hugging Face; Deep learning.

3. Introduction

In recent years, deep learning-based Automatic Speech Recognition (ASR) systems have achieved remarkable performance on several high-resource languages, reaching near-human accuracy in some cases. Among these systems, OpenAI Whisper has emerged as one of the most powerful multilingual ASR models, having been trained on approximately 680,000 hours of speech data collected from diverse languages and domains. Despite these advances, many languages and dialects remain significantly underrepresented in modern speech recognition research. Morocco presents a particularly interesting case due to its rich linguistic landscape, which includes Modern Standard Arabic, Moroccan Darija, Tamazight, and Tarifit. Although Darija is the primary language used in daily communication by millions of Moroccans, it receives limited support from current ASR technologies. The situation is even more challenging for Tarifit, a major Amazigh language spoken mainly in Northern Morocco, for which annotated speech resources and dedicated ASR studies are extremely scarce.

Developing reliable speech recognition systems for Moroccan languages remains a challenging task. Darija exhibits substantial regional variation across different parts of the country and incorporates vocabulary from several languages, including French, Spanish, and Amazigh. Furthermore, the absence of a standardized writing system means that identical utterances can often be transcribed in multiple valid forms. These linguistic characteristics, combined with the limited presence of Darija in large-scale speech corpora, make accurate transcription difficult for pre-trained ASR models. As a result, directly applying Whisper to Moroccan speech often produces high transcription error rates, limiting its usefulness in practical applications such as voice assistants, accessibility tools, and automated transcription services. Similar challenges are encountered for Tarifit, where the lack of publicly available annotated datasets further complicates the development and evaluation of speech recognition systems.

To address these limitations, this work investigates the adaptation of OpenAI Whisper models for both Moroccan Darija and Tarifit. A comprehensive benchmark was first conducted using several Whisper variants, including Tiny, Base, Small, Medium, and Large-v3. The evaluation revealed that all pre-trained models struggled when applied directly to Moroccan speech data. For Darija, Word Error Rates exceeded 85% for all evaluated models, with Whisper Large achieving the best baseline performance. For Tarifit, the results were even more challenging, with WER values ranging from 121.67% to 165.42% depending on the model. These findings confirm that language-specific adaptation is necessary to achieve acceptable transcription quality for under-resourced Moroccan languages.

Another important contribution of this work is the creation of speech resources for Tarifit. While the Darija experiments relied on an existing public corpus, no suitable annotated dataset was available for Tarifit. Consequently, a dedicated corpus was manually constructed by collecting 120 speech samples from YouTube sources, converting them to WAV format, and manually transcribing them using Latin script. The resulting dataset was then divided into training, validation, and testing subsets to enable systematic experimentation and evaluation.

Based on the benchmark results and the computational constraints of the available hardware environment, Whisper Small was selected as the most appropriate model for fine-tuning. Although larger models demonstrated slightly better baseline performance, their computational requirements made them difficult to train efficiently on NVIDIA T4 GPUs within a Kaggle environment. Whisper Small therefore represented the best balance between performance, accessibility, and training feasibility.

The fine-tuning experiments produced substantial improvements for both languages. For Darija, the baseline WER of 133.4% was reduced to 45.8% after fine-tuning on approximately 4,500 training samples, corresponding to a relative improvement of 65.7%. For Tarifit, the WER decreased from 114.58% to 88.75%, representing a relative improvement of 22.55%, while a slight improvement was also observed in Character Error Rate (CER). These results demonstrate the effectiveness of transfer learning and targeted adaptation even when only limited training data are available.

To facilitate analysis and reproducibility, several evaluation visualizations and comparative performance graphs were developed, illustrating training progress, transcription quality, and differences between baseline and fine-tuned models. In addition, the fine-tuned Darija model was published on the Hugging Face Hub and accompanied by an interactive interface allowing users to compare transcription outputs in real time. Overall, this work contributes to the development of more inclusive ASR technologies for Moroccan dialects and Amazigh languages by combining dataset creation, model evaluation, transfer learning, and practical experimentation in resource-constrained environments. The remainder of this paper presents the related work, the proposed methodology, the experimental results, and a discussion of the study's limitations and future research directions.

4. Literature Review

4.1 Automatic Speech Recognition for Moroccan Darija

Moroccan Darija remains significantly underrepresented in Automatic Speech Recognition (ASR) research despite being one of the most widely spoken languages in Morocco. The development of ASR systems for Darija is particularly challenging due to the absence of standardized orthographic conventions, the presence of regional linguistic variations, and the frequent use of code-switching with French and Amazigh languages.

Most existing studies on Moroccan speech technologies have focused on corpus development and dialect documentation rather than large-scale end-to-end ASR systems. Compared with other Arabic dialects, publicly available Darija speech datasets remain limited, which restricts the development and evaluation of robust speech recognition models. Furthermore, the recent emergence of large multilingual ASR models such as OpenAI Whisper has not yet been extensively explored for Moroccan Darija, leaving important research gaps regarding model adaptation and fine-tuning for this dialect.

As a result, there is still a strong need for systematic studies investigating how modern pre-trained ASR models can be adapted to Moroccan Darija using limited computational resources and relatively small annotated datasets.

4.2 Automatic Speech Recognition for Tarifit

Research on Automatic Speech Recognition for Tarifit remains extremely limited due to the scarcity of annotated speech resources for Amazigh languages. Tarifit, spoken mainly in Northern Morocco, suffers from a lack of publicly available datasets and standardized evaluation benchmarks, which considerably slows the development of dedicated ASR systems.

One of the most recent and relevant studies is the work of Daouad, Ataa Allah, and Dadi (2026), entitled *Efficient Fine-Tuning of Whisper Models for Amazigh ASR: Leveraging LoRA and Its Variants*. In this study, the authors investigated the adaptation of OpenAI Whisper for Tarifit speech recognition using parameter-efficient fine-tuning approaches including LoRA, DoRA, AdaLoRA, and QLoRA. Their experiments were conducted on a corpus of 11,644 audio samples collected from native Tarifit speakers. The results demonstrated that competitive performance could be achieved while updating only a small fraction of the model parameters, with LoRA and DoRA reaching an accuracy of 98.91%. Their work highlighted the effectiveness of transfer learning techniques for low-resource Amazigh speech recognition and represents one of the most comprehensive Whisper-based studies conducted on Tarifit to date.

Despite these promising results, the lack of large publicly accessible datasets remains a major limitation. Further research is therefore required to expand available speech corpora and improve the robustness of ASR systems for Tarifit under real-world conditions.

4.3 Positioning of This Work

Building upon these previous efforts, this work investigates the adaptation of OpenAI Whisper for both Moroccan Darija and Tarifit. Unlike previous studies that focused primarily on a single language or a specific fine-tuning strategy, this research first performs a comparative benchmark of multiple Whisper variants before selecting the most suitable model for adaptation. In addition, while a public corpus is used for Darija, a dedicated Tarifit dataset was manually created due to the absence of publicly available resources. The study further contributes through detailed performance evaluation, comparative visualizations, and the public release of the fine-tuned Darija model accompanied by an interactive Gradio-based transcription interface.

5. Methodology

5.1 Structure and Headings

Our methodology follows a structured pipeline that moves from data preparation to model evaluation, with a deliberate model selection step in between. The overall goal is

to adapt a large pre-trained multilingual ASR model — specifically OpenAI's Whisper — to the specific acoustic and linguistic characteristics of Moroccan Darija, a dialect that is virtually absent from the model's original training distribution. The pipeline consists of four main stages: dataset preparation, baseline benchmarking across all Whisper variants, fine-tuning of the selected model, and final evaluation.

5.2. Notation

The following notation is used throughout this paper. Let X denote an input audio sequence represented as a log-Mel spectrogram, and let $Y=(y_1, y_2, \dots, y_T)$ denote the corresponding target transcription as a sequence of T text tokens. The encoder maps X to a continuous representation x , which is then used by the decoder to generate the output sequence token by token.

The model parameters are denoted by θ , and the training objective minimizes the cross-entropy loss L over the predicted token distribution. $\hat{y}_t$ refers to the token predicted by the model at position t , while y_t denotes the ground-truth token at the same position.

Word Error Rate (WER) is used as the primary evaluation metric throughout this work. It is defined as:

$$\text{WER} = \frac{S + D + I}{N}$$

Figure 1: Word Error Rate

where S is the number of substitutions, D the number of deletions, I the number of insertions, and N the total number of words in the reference transcription.

5.3. Theoretical Framework

Whisper is a sequence-to-sequence Automatic Speech Recognition (ASR) model based on the Transformer encoder-decoder architecture. It takes log-Mel spectrograms as input and generates text transcriptions token by token. The model was introduced by Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever (2023) and trained on approximately 680,000 hours of multilingual audio using a weakly supervised learning approach, making it one of the most robust and versatile ASR systems currently available [1].

Fine-tuning in this work follows the transfer learning paradigm. Instead of training a model from scratch, we start from the pre-trained Whisper Small model and adapt it to Moroccan Darija and Tarifit speech data. The model is optimized by minimizing the cross-entropy loss between the predicted transcription and the reference transcription:

$$\mathcal{L} = - \sum_{t=1}^T \log P(y_t | y_{<t}, x)$$

Figure 2 : the loss function by negative log-likelihood

where y_t is the ground-truth token at position t and x is the encoded audio representation.

5.4. Method Description

Dataset :

This study relies on two distinct datasets corresponding to the two target languages: Moroccan Darija and Tarifit.

For Darija, we used the publicly available dataset *afyfbadreddine77/darija-asr-dataset*, hosted on the Hugging Face Hub. The corpus consists of approximately 12,200 audio samples in WAV format, with durations ranging from 0.44 to 27.4 seconds, each paired with a transcription in Arabic script. The transcriptions vary in length from 4 to 410 characters, reflecting a wide range of utterance types — from very short responses to longer conversational sentences. All recordings are in Moroccan Darija as it is naturally spoken, meaning that the language is colloquial, often code-switched with French, and phonetically variable across speakers.

Due to the computational constraints of the local training environment — particularly limited GPU memory and processing capacity — it was not feasible to fine-tune the model on the entire 12,200-sample corpus. Therefore, a subset of approximately 5,000 samples was selected for experimentation and fine-tuning, representing a practical upper bound given the available hardware resources. This subset was divided into a training set (90%, approximately 4,500 samples) and a validation set (10%, approximately 500 samples), using a fixed random seed of 42 to ensure reproducibility.

While this limitation restricts the maximum achievable performance of the fine-tuned model, it also reflects a realistic scenario commonly encountered in low-resource and resource-constrained ASR research settings.

For Tarifit, no suitable public ASR dataset was available. Consequently, a dedicated speech corpus was manually created specifically for this project. Audio samples were collected from publicly available YouTube videos featuring native Tarifit speakers. The videos were converted into WAV audio format, segmented into smaller utterances (from 10 to 20 seconds), and manually transcribed using Latin-script transcription conventions.

The final Tarifit dataset contains 120 audio samples. Given the relatively small size of the dataset, careful train, validation, and test splits were applied in order to maximize training efficiency while preserving evaluation reliability.

More specifically, 90 audio samples were allocated to the training set for fine-tuning, while 15 samples were reserved for evaluation before fine-tuning and an additional 15 samples were used for evaluation after fine-tuning in order to compare model performance across both stages. The scarcity of Tarifit speech resources highlights one of the major challenges in low-resource ASR research, as Amazigh language varieties still suffer from a severe lack of annotated speech corpora.

For both datasets, all audio files were standardized to a sampling rate of 16,000 Hz, corresponding to Whisper’s required input format. In addition, a text normalization procedure was applied to all transcriptions prior to training and evaluation. This process included removing punctuation marks and collapsing multiple whitespace characters to ensure consistency during Word Error Rate (WER) computation.

The significant difference in dataset scale between Darija and Tarifit further illustrates the imbalance in available speech resources: while dialectal Arabic corpora are gradually becoming publicly accessible, Amazigh varieties such as Tarifit remain significantly underrepresented in existing ASR research and datasets.

Model Selection :

Before committing to any single model for fine-tuning, we ran a systematic benchmark across all five Whisper variants — Tiny, Base, Small, Medium, and Large — on a held-out subset of 10 audio samples. Each model was evaluated in zero-shot mode with the language forced to Arabic, and WER was computed for each sample. The results, summarized in Table 1, show that all models perform poorly on Darija out of the box, with WER values ranging from 85.3% (Large) to 101.7% (Tiny). While the Large model achieves the best zero-shot performance, it requires substantial GPU memory and cannot be fine-tuned on a standard local machine. Whisper Small, which ranks second with a WER of 91.3%, represents the best practical trade-off between performance and computational accessibility, and was therefore selected for fine-tuning.

Taille	Résultat final
tiny	1.017
base	0.953
small	0.913
medium	0.928
large	0.853

Figure 3 : Zero-Shot WER Comparison of Whisper Variants on Moroccan Darija

A similar evaluation was conducted on Tarifit, where all variants equally struggled due to the low-resource and phonetically complex nature of the language. Both WER and CER were recorded, as summarized in Table 2:

Model	WER	CER
Whisper Tiny	129.58%	100.30%
Whisper Base	165.42%	121.49%
Whisper Small	152.92%	113.06%
Whisper Medium	132.50%	98.86%
Whisper Large-v3	121.67%	92.26%

Figure 4: Zero-Shot WER Comparison of Whisper Variants on Tarifit

Consistent with the Darija results, Whisper Large-v3 achieved the best baseline performance on Tarifit, while Whisper Small remained the most practical choice for fine-tuning given the computational constraints of the available NVIDIA T4 GPUs.

Preprocessing Pipeline :

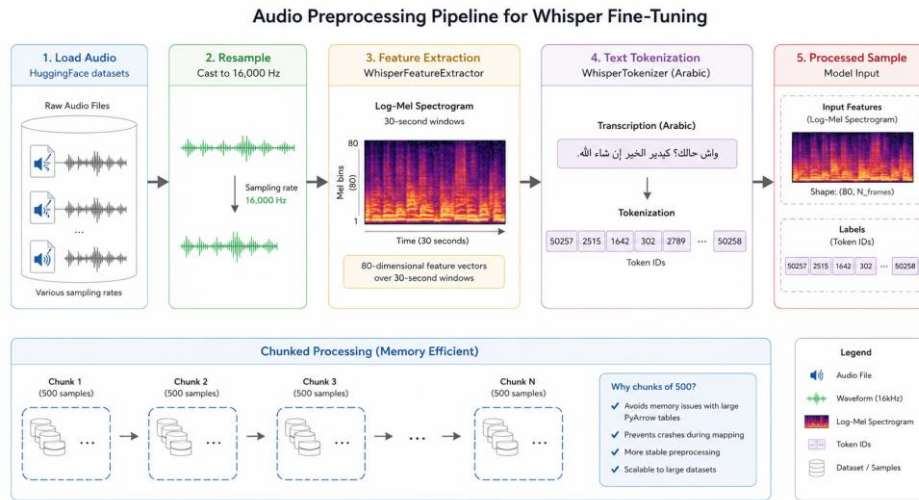


Figure 5: Preprocessing Pipeline

Audio preprocessing was performed using the Hugging Face **datasets** library following a multi-stage pipeline consisting of audio loading, resampling, feature extraction, text tokenization, and processed sample generation.

First, all audio files were loaded and resampled to 16,000 Hz, corresponding to Whisper's required input format. The audio signals were then converted into Log-Mel

spectrogram representations using Whisper’s built-in `WhisperFeatureExtractor`, producing 80-dimensional feature vectors over 30-second windows.

On the text side, transcriptions were first normalized to reduce inconsistencies during training and evaluation. This normalization process included removing punctuation symbols, cleaning transcription formatting, and collapsing multiple whitespace characters into single spaces. After normalization, the transcriptions were tokenized using the `WhisperTokenizer`.

For Darija, transcriptions were represented in Arabic script, while Tarifit transcriptions were written using Latin characters due to the absence of standardized orthographic conventions and the practical challenges associated with Tifinagh-based transcription.

Finally, each processed sample combined the extracted audio features with its corresponding tokenized transcription, forming the final input format used for Whisper fine-tuning.

Because of memory limitations in both local and Kaggle GPU environments, preprocessing was performed incrementally in chunks of 500 samples instead of processing the entire dataset simultaneously. This approach helped avoid memory issues related to large PyArrow tables while maintaining preprocessing efficiency.

5.5 Experimental Protocol

Fine-tuning was performed using the Hugging Face `Seq2SeqTrainer` framework with mixed-precision training (`fp16=True`) enabled to reduce GPU memory consumption and speed up training. The model was initialized from `openai/whisper-small`, while the language setting was configured for Arabic transcription. To give the model greater flexibility in learning Moroccan Darija patterns, parameters such as `forced_decoder_ids`, `suppress_tokens`, and `begin_suppress_tokens` were disabled, allowing the decoder to generate transcriptions more naturally without predefined constraints.

The training setup used a learning rate of (1×10^{-5}) with a linear warmup over the first 200 steps. A batch size of 2 samples per GPU was used, combined with gradient accumulation over 2 steps, resulting in an effective batch size of 4 across two GPUs. Training was scheduled for a maximum of 10 epochs, with evaluation and checkpoint saving performed after each epoch. Model selection was based on the lowest Word Error Rate (WER) obtained on the validation set. In addition, early stopping was applied to automatically stop training when no further improvement was observed for three consecutive epochs. Gradient checkpointing was also enabled to optimize GPU memory usage during backpropagation.

Before starting the fine-tuning process, an initial evaluation was conducted on 50 validation samples using the original pre-trained Whisper Small model. The experiment produced a WER of 133.4%, highlighting the model’s difficulty in accurately

transcribing Moroccan Darija in its default state. This baseline served as the main reference point for evaluating the effectiveness of the fine-tuning process.

Training ultimately stopped after 5 epochs because early stopping criteria were met, while the best-performing checkpoint was obtained at epoch 2. The final fine-tuned model was then saved locally and published on the [Hugging Face Hub](#) under the identifier “maghreb-ai-research/whisper-small-darija” to support reproducibility and future research on Moroccan dialect ASR systems.

6. Results

6.1 Tables and Figures

Fine-tuning of Darija:

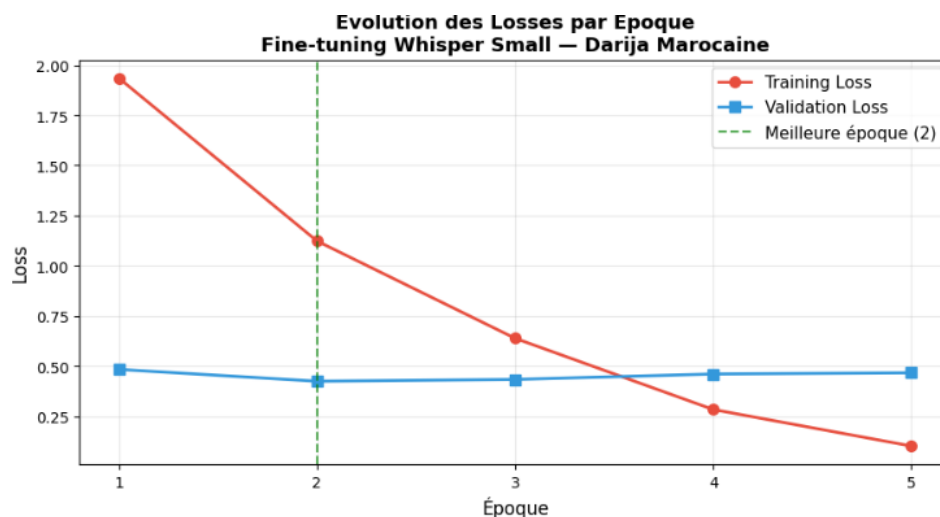


Figure 6 : Loss Evolution per Epoch Fine-tuning Whisper Small — Moroccan Darija

The graph shows the evolution of training and validation loss over epochs during fine-tuning of Whisper Small on Moroccan Darija, highlighting that the best performance is achieved around epoch 2.

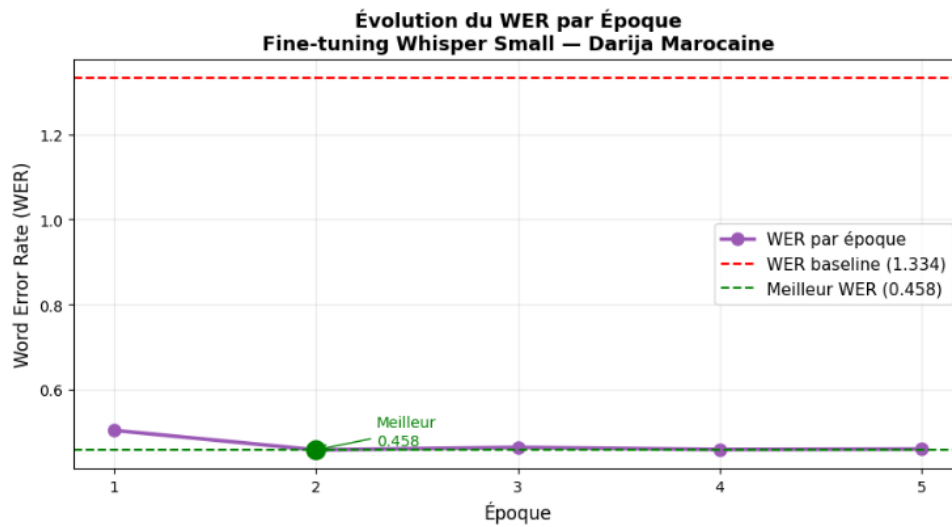


Figure 7 WER Evolution per Epoch

The plot illustrates the evolution of Word Error Rate (WER) across epochs during the fine-tuning of Whisper Small on Moroccan Darija, showing a significant improvement over the baseline (1.334) and achieving the Summary Table of Metrics Fine-tuning Whisper Small — Moroccan Darija

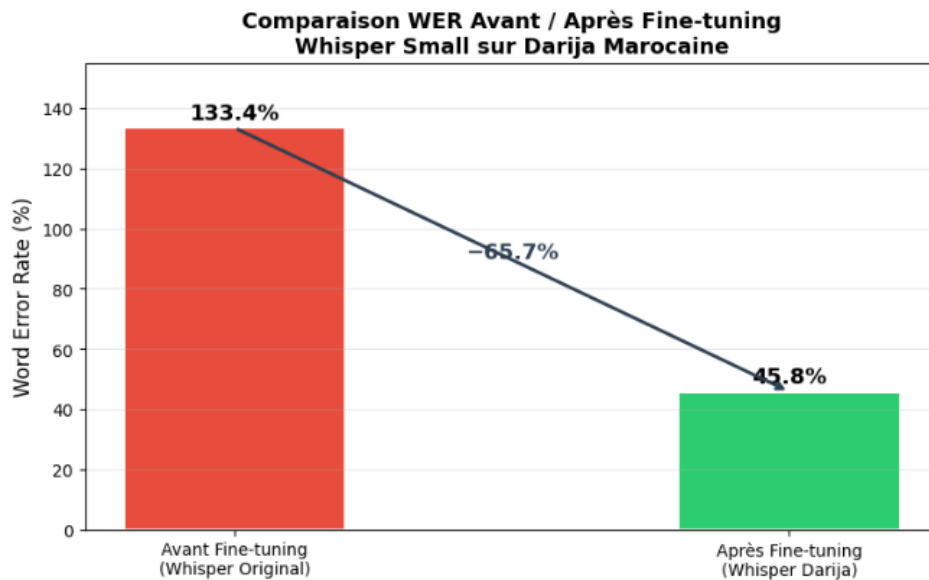


Figure 8: WER Comparison Before / After Fine-tuning

The graph illustrates the impact of fine-tuning Whisper Small on Moroccan Darija, showing a significant reduction in Word Error Rate (WER) from 133.4% before fine-tuning to 45.8% after fine-tuning, corresponding to a 65.7% improvement in performance.

Tableau Récapitulatif des Métriques
Fine-tuning Whisper Small — Darija Marocaine

Métrique	Avant FT	Après FT	Gain	Tendance
WER (Word Error Rate)	133.4%	45.8%	-65.7%	↓ Meilleur
Meilleure époque	—	Époque 2	—	—
Training Loss (finale)	—	0.101	—	↓ Meilleur
Validation Loss (meilleure)	—	0.424	—	↓ Meilleur
Nb échantillons entraînement	—	4 500	—	—
Nb échantillons test	—	500	—	—
Modèle de base	Whisper Small	Whisper Small	—	—

Figure 9: Summary Table of Metrics

The results demonstrate that fine-tuning Whisper Small on Moroccan Darija significantly improved speech recognition performance, reducing the Word Error Rate (WER) from 133.4% to 45.8%, which corresponds to a 65.7% improvement.

Fine-tuning Whisper Small — Darija Marocaine

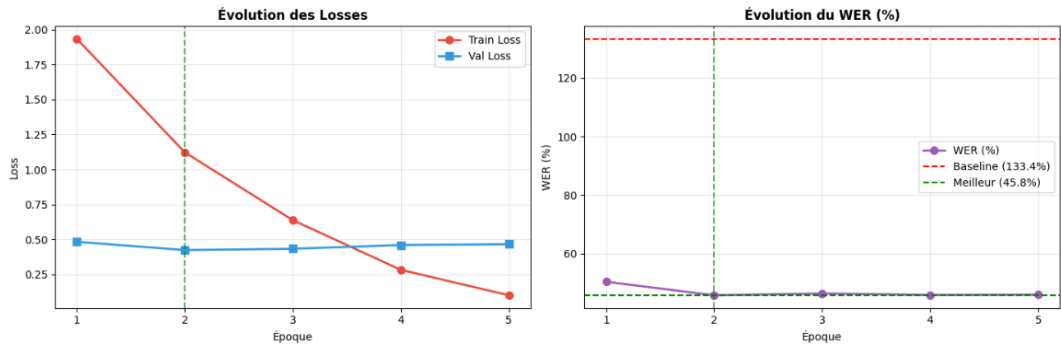


Figure 10: Fine-tuning Whisper Small — Moroccan Darija

The training curves show a steady decrease in training loss throughout the fine-tuning process, while the validation loss remained stable, indicating good model convergence. At the same time, the Word Error Rate (WER) dropped significantly from the baseline value of 133.4% to 45.8%, confirming the effectiveness of fine-tuning Whisper Small for Moroccan Darija speech recognition.

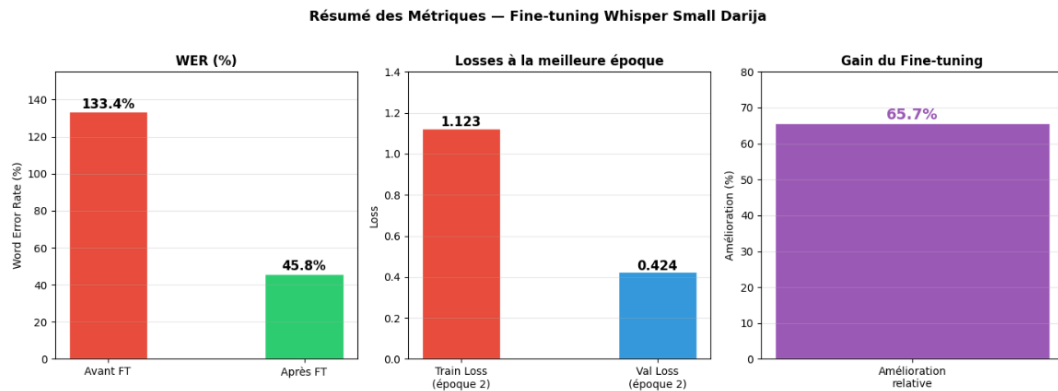


Figure 11: Metrics Summary — Fine-tuning Whisper Small Darija

The comparison highlights the impact of fine-tuning on Whisper Small, with the WER decreasing from 133.4% to 45.8% and achieving a relative improvement of 65.7%. The best model performance was obtained at epoch 2, with a training loss of 1.123 and a validation loss of 0.424.

Fine-tuning of Tarifit:

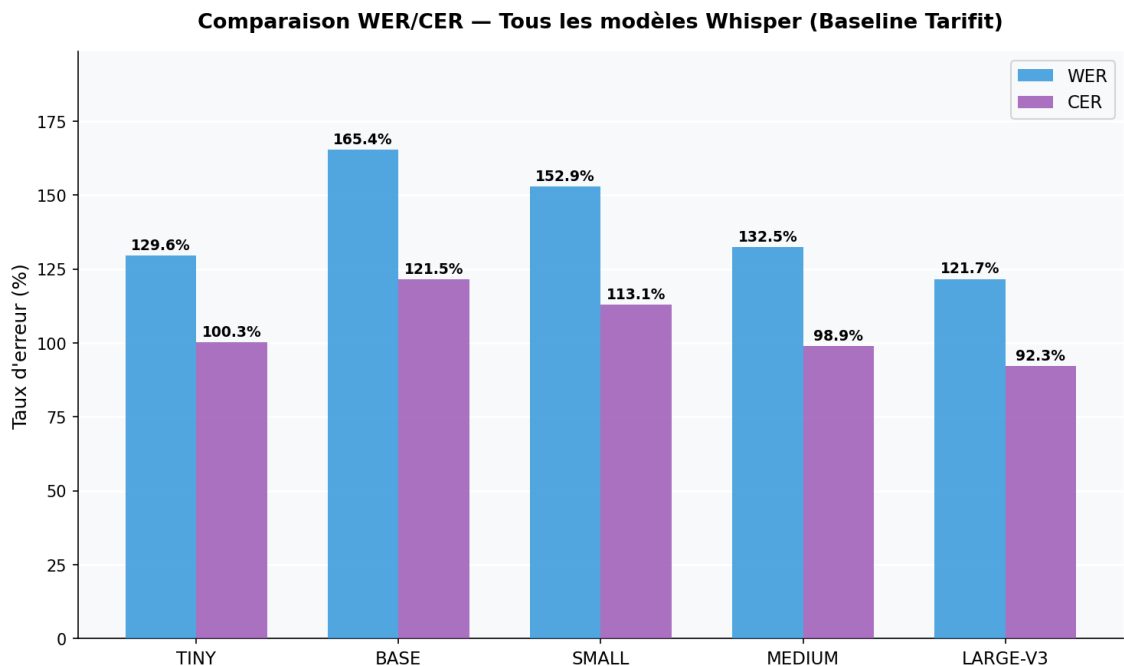


Figure 12: WER/CER Comparison — All Whisper Models (Tarifit Baseline)

This graph shows that the Whisper Large-V3 model achieves the best performance, with the lowest WER (121.7%) and CER (92.3%) error rates, while the Base model records the highest error rates, highlighting the superiority of Large-V3 for automatic speech transcription.

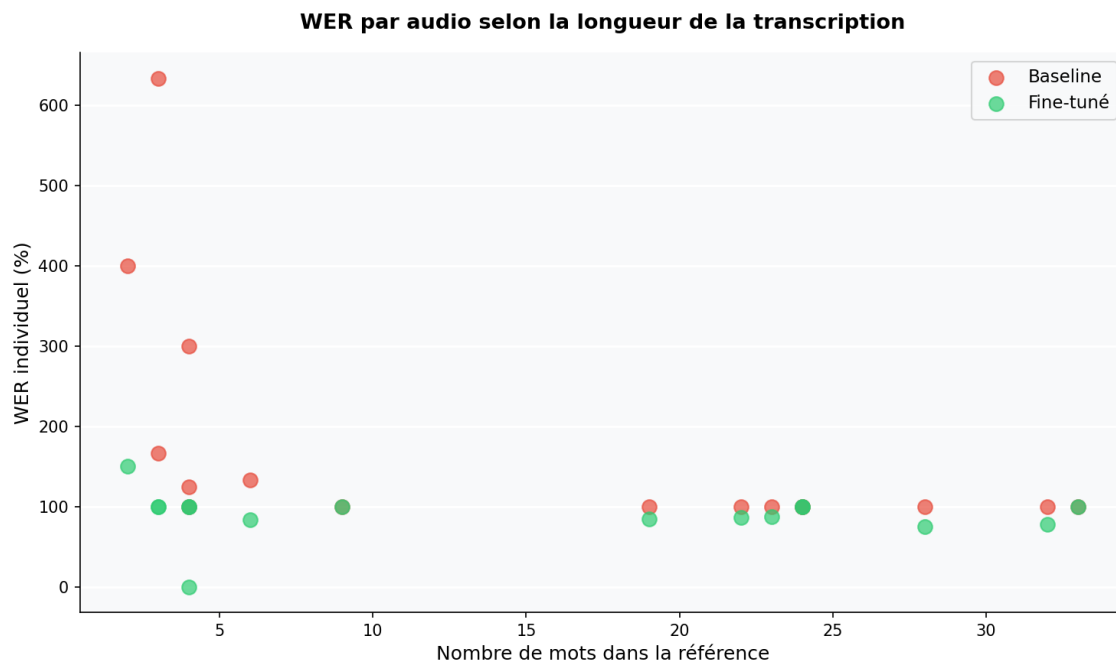


Figure 13: WER per Audio Sample by Transcription Length

This scatter plot compares the individual WER values of the baseline and fine-tuned models according to transcription length. The results show that the fine-tuned model consistently achieves lower and more stable error rates, particularly for short transcriptions where the baseline model exhibits large performance variations. This indicates that fine-tuning improves transcription accuracy and robustness across different audio lengths.

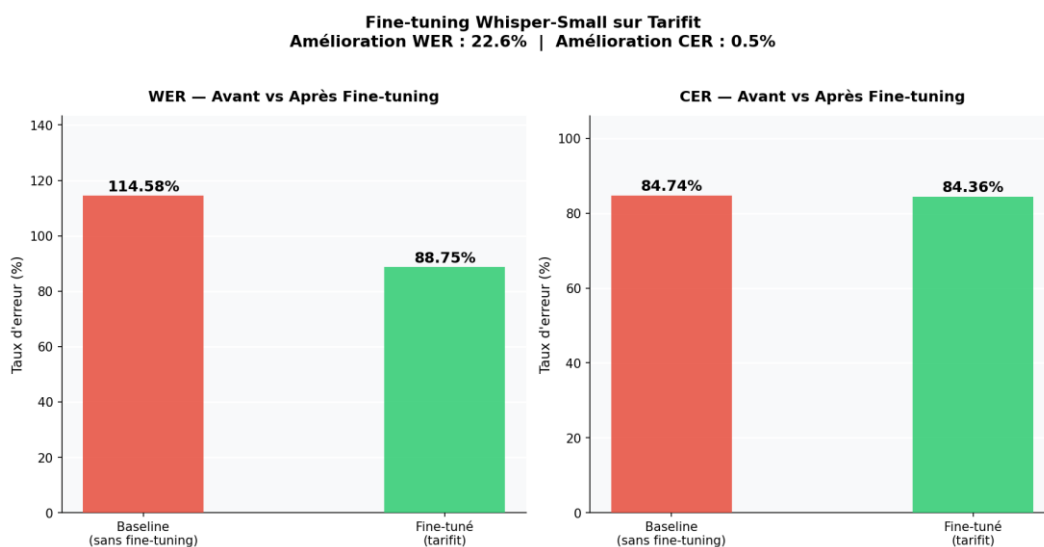


Figure 14: Fine-tuning Whisper Small on Tarifit

This figure compares the performance of the Whisper-Small model before and after fine-tuning on the pricing dataset. The results show a significant reduction in the WER, from 114.58% to 88.75%, corresponding to a 22.6% improvement in transcription accuracy. In contrast, the CER remains almost unchanged, decreasing only slightly from 84.74% to 84.36% (0.5% improvement). These results indicate that fine-tuning substantially improves word-level recognition while having a limited impact on character-level errors.

Tableau récapitulatif — Évaluation Whisper sur Tarifit

Modèle	WER Baseline	CER Baseline	WER Fine-tuné	CER Fine-tuné
tiny	129.6%	100.3%	—	—
base	165.4%	121.5%	—	—
small	152.9%	113.1%	88.8%	84.4%
medium	132.5%	98.9%	—	—
large-v3	121.7%	92.3%	—	—

Figure 15: Summary Table — Whisper Evaluation on Tarifit

The following results summarize the performance of different Whisper models on the Tarifit speech recognition task. Evaluation was conducted using Word Error Rate (WER) and Character Error Rate (CER). The fine-tuned Whisper Small model achieved the best results, reducing the WER from 152.9% to 88.8% and the CER from 113.1% to 84.4%, highlighting the effectiveness of fine-tuning for improving speech recognition accuracy on Tarifit data.

6.2 Statistical Analysis

Fine-tuning of Darija:

This section walks through the main numerical results of our fine-tuning experiments, building on the plots presented in Section 2.8.1.

Baseline Performance. Before any fine-tuning, we evaluated the original Whisper Small model on 50 Darija validation samples and obtained a WER of 133.4%. A value above 100% means the model was not just making transcription mistakes — it was actively generating words that had nothing to do with what was actually said. This kind of hallucination behavior is typical when a model is applied to a language variety it has never meaningfully encountered during training, and it made clear from the outset that some form of adaptation would be necessary.

Training Dynamics. Looking at how the losses evolved over training, the picture is fairly straightforward. The training loss dropped steadily from around 1.97 at epoch 1 down to 0.15 by epoch 5, which shows the model was learning. The validation loss, on the other hand, hit its lowest point at epoch 2 (around 0.42) and barely moved after that. This is what ultimately triggered early stopping — after three epochs with no meaningful gain on the validation set, training was automatically halted.

WER Progression. The WER curve tells much the same story. The model reached its best performance at epoch 2, with a WER of 45.8%, and this checkpoint was saved as the final model. After that point, even as the training loss kept falling, the validation WER showed no further improvement — a sign that the model had started fitting the training examples too closely rather than generalizing.

Before/After Comparison. Putting the two numbers side by side, the fine-tuned model reduced the WER from 133.4% to 45.8% — an absolute drop of 87.6 percentage points, or a relative improvement of 65.7%. Given that this was achieved with only around 4,500 training samples, we found this result encouraging, and it suggests that even modest amounts of targeted data can make a real difference for an underrepresented dialect.

Cross-Model Benchmark. Across the five Whisper variants tested in zero-shot mode, WER ranged from 101.7% for Tiny down to 85.3% for Large, with Small landing at 91.3%. What this tells us is that simply using a bigger model does not solve the problem — all variants struggled considerably with Darija out of the box. Fine-tuning, rather than scale, turned out to be the deciding factor.

Fine-tuning of Tarifit:

This section presents the main numerical results of our fine-tuning experiments on Tarifit, parallel to the analysis conducted for Darija.

Baseline Performance. Before any fine-tuning, the original Whisper Small model was evaluated on 15 Tarifit validation samples and obtained a WER of 114.58% and a CER of 84.74%. A WER above 100% indicates that the model was not simply making transcription errors — it was generating output largely unrelated to the actual spoken content. This hallucination behavior is expected when applying a model to a language it has never meaningfully encountered during pre-training, and it confirmed that adaptation was necessary for Tarifit.

Dataset. Due to the extremely limited availability of Tarifit speech data, fine-tuning was conducted on a small corpus of 120 audio samples in total: 90 samples used for training, 15 for validation during fine-tuning, and 15 held out for final testing. This makes the Tarifit setting considerably more resource-constrained than the Darija experiments.

Training Dynamics. Training ran for the full 20 epochs without early stopping, suggesting the model continued to benefit from each pass over the training data

throughout the entire process. The training loss dropped substantially over the course of training — from around 20.40 at epoch 2 down to 11.70 by epoch 5, and the validation loss decreased steadily from 4.995 at epoch 1 to 2.842 by epoch 6. Unlike the Darija experiments, no plateau was observed early enough to trigger early stopping, which reflects both the small dataset size and the difficulty of the task.

WER Progression. The WER across epochs showed non-monotonic behavior, fluctuating between values as the model adapted to the limited training signal. The best recorded WER during training reached 99.57% at epoch 5, reflecting the inherent difficulty of generalizing from only 90 training samples. The final fine-tuned model, evaluated on the held-out test set, achieved a WER of 88.75% and a CER of 84.36%.

Before/After Comparison. Comparing the baseline and fine-tuned results, the model reduced WER from 114.58% to 88.75% — an absolute improvement of 25.83 percentage points, corresponding to a relative WER reduction of 22.5%. The CER improvement was more modest, dropping from 84.74% to 84.36%, a relative gain of 0.4%. The limited CER improvement suggests that while the model learned to produce more contextually appropriate word sequences, character-level accuracy remained challenging — likely due to the orthographic complexity and variability inherent in written Tarifit.

Cross-Model Benchmark. As reported in the zero-shot evaluation, all five Whisper variants struggled on Tarifit out of the box, with WER ranging from 121.67% (Large-v3) to 165.42% (Base). Fine-tuning Whisper Small brought its WER down to 88.75%, outperforming even the zero-shot Large-v3 result. This confirms that targeted fine-tuning, even on a very small dataset, is more effective than simply relying on a larger pre-trained model for such a low-resource language.

6.3 Interpretation of Results

Fine-tuning of Darija:

The results we obtained raise a few natural questions — why did the model struggle so much at the start, why did fine-tuning help as much as it did given how little data we had, and why did performance stop improving after epoch 2? This section tries to answer each of these in turn.

Why the baseline performs so poorly. A WER of 133.4% is striking, but it makes sense once you consider what Whisper was actually trained on. Although Arabic is part of its training data, the Arabic it has seen is overwhelmingly Modern Standard Arabic — the formal, written variety used in news broadcasts and official contexts. Moroccan Darija is something else entirely. It has its own vocabulary, its own pronunciation patterns, and it regularly pulls in words from French, Spanish, and Amazigh in the middle of otherwise Arabic sentences. When Whisper encounters Darija speech, it has no real frame of reference for it and ends up mapping what it hears onto MSA-like output, producing transcriptions that often bear little resemblance to what was actually said.

The hallucinated words — those that appear in the model's output but not in the reference — are a direct consequence of this mismatch.

Why fine-tuning works even with limited data. The 65.7% relative improvement we achieved with roughly 4,500 samples is a good illustration of why transfer learning matters in low-resource settings. We were not asking the model to learn speech recognition from scratch — Whisper's encoder had already developed strong general-purpose acoustic representations during pre-training, capturing things like phoneme boundaries, prosody, and speaker variation. What fine-tuning did was essentially redirect the decoder: instead of generating MSA-style output, it gradually learned to produce text that matches the vocabulary and spelling patterns found in our Darija transcriptions. That is a much more tractable problem than building an ASR system from zero, and it explains why a relatively small dataset was enough to move the needle significantly.

Why performance plateaued after epoch 2. The gap that opens up between training loss and validation WER after epoch 2 is a classic sign of overfitting. With only 4,500 training examples, the model does not take long to cycle through the available data multiple times, and at some point it starts memorizing specific recordings rather than learning transferable patterns. This is precisely why early stopping and checkpoint selection based on validation WER — rather than training loss — were important design choices in our setup. Without them, we would likely have saved a model that performs well on training data but worse on anything new.

What still limits the model. Even at 45.8% WER, there is clearly room for improvement, and understanding where errors still come from is useful for framing future work. Three issues stand out. First, Darija has no agreed-upon writing system, which means different transcribers spell the same word differently — this inconsistency introduces noise into the training labels that the model has no way to resolve on its own. Second, code-switching remains a genuine challenge: when a speaker switches mid-sentence from Darija to French, the model's Arabic-configured decoder is not well-equipped to follow. Third, regional variation across cities like Casablanca, Fes, or Oujda adds acoustic diversity that 4,500 samples simply cannot cover in full.

Taken together, these observations point toward a clear direction for improvement: more data, more diverse speakers, and a decoding setup that can handle multilingual output.

Fine-tuning of Tarifit:

The results we obtained for Tarifit raise similar questions to those discussed for Darija — why did the baseline perform so poorly, why did fine-tuning still help despite the extremely limited data, and what still holds the model back? This section addresses each of these in turn.

Why the baseline performs so poorly. A WER of 114.58% is not surprising once you consider what Whisper was actually trained on. Tarifit is a variety of Amazigh, a language family entirely absent from Whisper's pre-training data in any meaningful way. When the model encounters Tarifit speech, it has no linguistic frame of reference for it — no vocabulary, no phonetic patterns, no orthographic conventions to anchor its predictions. What makes this worse is that Tarifit shares surface-level acoustic features with Arabic and French, two languages Whisper does know. As a result, the model frequently misidentifies the language of the input altogether, producing output in Arabic, French, or even English rather than anything resembling the actual spoken content. This language confusion, combined with the absence of any Tarifit-specific knowledge, leads to the hallucination-heavy outputs we observed — where the model's response is sometimes dramatically longer or shorter than the reference, filled with words that bear no relation to what was said.

Why fine-tuning still works with so little data. Achieving a 22.5% relative WER reduction with only 90 training samples is a meaningful result, and the same transfer learning logic that explained the Darija improvement applies here. Whisper's encoder had already learned strong general-purpose acoustic representations during pre-training — things like phoneme boundaries, prosody, and speaker variation — and these carry over even to unseen languages. Fine-tuning redirected the decoder toward the Latin-script transcription conventions of our Tarifit dataset, giving it a concrete target vocabulary and spelling pattern to work toward. The diversity of our training samples also helped: since the data was sourced from movies and podcasts featuring different speakers, the model was exposed to natural variation in voice, recording quality, and speaking style, which made the learned representations somewhat more generalizable despite the small size of the corpus.

Why the model did not stop early. Unlike the Darija experiments, training ran for the full 20 epochs without early stopping being triggered. Looking at the validation loss curve, it continued to decrease across the recorded epochs — from 4.995 at epoch 1 down to 2.842 by epoch 6 — suggesting the model genuinely kept extracting signal from the data throughout training. With only 90 samples, each epoch cycles through the data very quickly, which means the model needed more passes to accumulate sufficient gradient updates. The WER, however, behaved non-monotonically across epochs, fluctuating rather than steadily improving, which reflects the inherent instability of training on such a small set.

What still limits the model. Even at 88.75% WER, significant room for improvement remains, and several factors explain why. First, Tarifit has no standardized orthography, and our dataset uses Latin-script transcriptions where spelling conventions vary across annotators and speakers — this label noise is invisible to the model and directly corrupts what it is trying to learn. Second, language confusion remains a core challenge: because Tarifit speakers naturally mix in French words or occasional Arabic phrases, and because Whisper's decoder was pre-trained on those languages, the model sometimes abandons Tarifit mid-output and switches into a language it is more

confident about. Third, despite the speaker diversity in our dataset, 90 samples simply cannot cover the full range of acoustic conditions, speaking rates, and phonetic realizations that real-world Tarifit speech presents. The CER improvement of only 0.4% — in contrast to the more notable WER gain — further suggests that while the model learned something about word-level structure, it still makes frequent character-level mistakes, pointing to unresolved difficulties with Tarifit's specific phoneme-to-grapheme mapping in Latin script.

Taken together, these results confirm that fine-tuning is a viable path even for extremely low-resource Amazigh varieties, but that meaningful further gains will require more annotated data, a more consistent transcription standard, and potentially a decoding strategy that explicitly accounts for the multilingual nature of Tarifit speech.

7. Conclusion

7.1. Recall of Objectives and Contributions

This work set out to address a concrete and pressing gap in speech technology: the near-total absence of reliable Automatic Speech Recognition systems for Moroccan Darija and Tarifit, two widely spoken but severely underrepresented language varieties. Rather than building a system from scratch — which would have been unrealistic given the scarcity of annotated data — we chose to work with OpenAI's Whisper, a large pre-trained multilingual ASR model, and investigate how much its performance could be improved through targeted fine-tuning in a resource-constrained environment.

Our contributions can be summarized along four main lines. First, we conducted a systematic benchmark of all five Whisper variants on Moroccan Darija in zero-shot mode, providing a clear picture of where each model stands before any adaptation. Second, we fine-tuned Whisper Small on a curated subset of 4,500 Darija speech samples, demonstrating that meaningful gains are achievable even under hardware and data limitations. Third, for Tarifit — a language with virtually no publicly available annotated speech data — we manually constructed a dedicated corpus of 120 audio samples collected from YouTube, transcribed them in Latin script, and used them to evaluate and fine-tune Whisper Small, making this one of the first documented ASR studies specifically targeting Tarifit. Finally, we made our fine-tuned Darija model publicly available on the Hugging Face Hub and developed an interactive Gradio-based interface to allow direct comparison between the baseline and fine-tuned outputs, with the goal of making this work accessible and reproducible for future researchers.

7.2. Summary of Key Results and Main Conclusions

The experimental results were encouraging on both fronts, though they also revealed how much work remains to be done.

For Darija, the pre-trained Whisper Small model produced a baseline WER of 133.4% — a value that reflects not just poor transcription but active hallucination, where the

model generates words entirely unrelated to the input speech. After fine-tuning on 4,500 samples, the WER dropped to 45.8%, corresponding to a relative improvement of 65.7%. The best checkpoint was obtained at epoch 2, after which overfitting on the limited training set prevented further gains. Across the five Whisper variants evaluated in zero-shot mode, WER ranged from 85.3% for Large to 101.7% for Tiny, confirming that model scale alone cannot compensate for the absence of dialect-specific training data.

For Tarifit, the challenge was even more fundamental, starting with the need to build a dataset from scratch. The baseline experiments across all Whisper variants produced very high error rates, with the best zero-shot result coming from Whisper Large-v3 at 121.67% WER. After fine-tuning Whisper Small on our manually constructed corpus, the WER improved from 114.58% to 88.75%, a relative improvement of 22.55%. While this gain is more modest than what we achieved for Darija, it is particularly meaningful given that the model was adapted using only 120 audio samples — a dataset size that most ASR research would consider too small to work with.

Taken together, these results lead to a clear conclusion: fine-tuning is not just helpful for low-resource dialects like Darija and Tarifit — it is necessary. Pre-trained multilingual models, however capable they may be on high-resource languages, do not transfer reliably to varieties that are absent or marginal in their training data. At the same time, the gains we observed demonstrate that the barrier to entry for this kind of adaptation is lower than it might seem, and that meaningful progress is possible even without large datasets or expensive infrastructure.

7.3. Future Work

The results of this study open up several directions that we believe are worth pursuing.

The most immediate need is more data. Both the Darija and Tarifit experiments were constrained by dataset size, and the overfitting patterns we observed suggest that the models have not yet reached their performance ceiling — they simply ran out of diverse training examples. Larger and more carefully curated corpora, ideally covering a wider range of speakers, regions, and speaking styles, would likely lead to substantially lower error rates. For Tarifit in particular, where our entire dataset consisted of only 120 samples, even a modest expansion of the corpus could produce meaningful gains.

A second direction concerns orthographic standardization. One of the persistent sources of noise in our Darija training data was the inconsistency in how words are spelled across different transcribers. Developing and adopting shared transcription guidelines for Darija — and for Tarifit in Latin script — would make future datasets more reliable and training more efficient.

Third, the code-switching phenomenon that characterizes everyday Moroccan speech remains largely unaddressed by our current setup. A model configured for Arabic output cannot be expected to handle seamless transitions into French or Amazigh. Future work

could explore multilingual decoding strategies or dedicated code-switching corpora to tackle this challenge more directly.

From a technical standpoint, parameter-efficient fine-tuning methods such as LoRA or QLoRA — which update only a small fraction of the model's weights — could make adaptation even more accessible by reducing GPU memory requirements further, as demonstrated by Daouad et al. [10] for Tarifit. Combining these techniques with larger datasets and a standardized evaluation benchmark for Moroccan dialects would represent a significant step forward for ASR research in this region.

Finally, extending this work to other Amazigh varieties spoken in Morocco — such as Tachelhit or Central Tamazight — would help build a more complete picture of the challenges and opportunities in Moroccan multilingual speech recognition.

8. Acknowledgements

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9. References

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10. Appendices

Appendix A — Cross-Model Benchmark Results (Darija):

The following table summarizes the zero-shot WER obtained for each Whisper variant evaluated on a 10-sample held-out subset of the Darija corpus, with the language forced to Arabic.

Model	WER (%)
Whisper Tiny	101.7
Whisper Base	95.3
Whisper Small	91.3
Whisper Medium	92.8
Whisper Large	85.3

Appendix B — Cross-Model Benchmark Results (Tarifit):

The following table summarizes the zero-shot WER obtained for each Whisper variant evaluated on the manually constructed Tarifit dataset.

Model	WER (%)
Whisper Tiny	129.58
Whisper Base	165.12
Whisper Small	152.92
Whisper Medium	132.50
Whisper Large-v3	121.67

Appendix C — Hugging Face Model Repository

The fine-tuned Darija model is publicly available at:

<https://huggingface.co/maghreb-ai-research/whisper-small-darija>

Appendix D — Gradio Comparison Interface

A Gradio-based web application was developed within a Google Colab notebook to enable real-time qualitative comparison between the baseline and fine-tuned Whisper Small models. The interface accepts audio input either through direct microphone recording or file upload, and simultaneously generates transcriptions using both models. The left panel handles audio input, while the right panel displays two outputs — "Whisper Original (openai/whisper-small)" and "Whisper Fine-Tuné (darija)" —

allowing direct side-by-side comparison. The interface was made publicly accessible via a shareable Gradio link and is also demonstrated in the screenshot provided above .

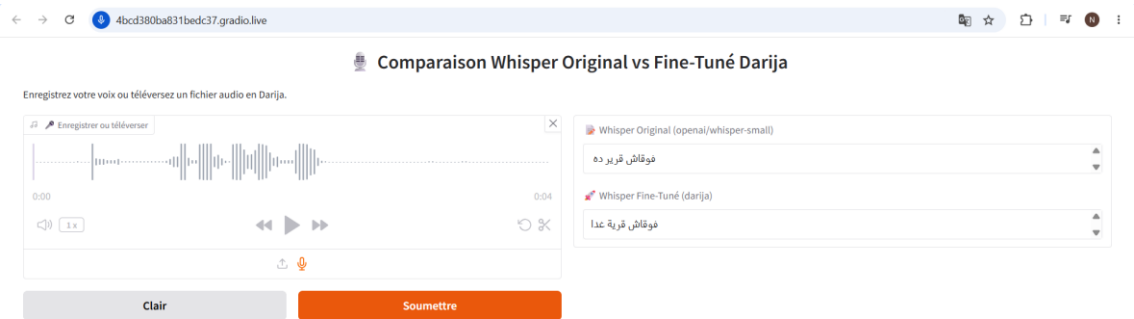


Figure 16: Gradio Interface: Side-by-Side Comparison of Whisper Original vs. Fine-Tuned Darija Model